FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

CHEMISTRY

Past Paper 2024 • Paper Code: 2HA-I 24005 — 40054 (Version B) / 42052 (Version D)

Retyped faithfully from the original board paper — same questions, no changes.

VERSION B

CHEMISTRY HSSC-II — SECTION – A (Objective Type)

Time Allowed: 25 Minutes | **Maximum Marks:** 17 | **Paper Code:** 40054 (Version B)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. Reducing smog contains high concentration of:

(A) NO2
(B) O3
(C) PAN
(D) SO2

Q2. In UV-Vis spectroscopy, n → π* transition occurs in:

(A) Ethene
(B) Ethanol
(C) Ethanal
(D) Ethane

Q3. Which one is NMR inactive nucleus?

(A) 1H
(B) 15N
(C) 16O
(D) 13C

Q4. In which of the following pairs both elements have no tendency to form dichlorides?

(A) C and Sn
(B) Ge and C
(C) Sn and Si
(D) C and Si

Q5. MgCO3 is NOT thermally stable due to:

(A) Less electropositivity of Mg
(B) High charge density of Mg+2
(C) Low charge density of Mg+2
(D) Less electronegativity of Mg

Q6. Which of the following elements will react with O2 to form an amphoteric oxide?

(A) Mg
(B) Al
(C) Si
(D) Na

Q7. Identify the colourless complex ion:

(A) [Fe(H2O)6]2+
(B) [Zn(NH3)4]2+
(C) [Cu(NH3)4]2+
(D) [Cr(H2O)6]3+

Q8. Identify the functional group in the compound formed by the reaction of an alcohol with a carboxylic acid:

(A) -C(=O)-OR
(B) -C(=O)-
(C) -C(=O)-H
(D) -C(=O)-O-C(=O)-

Q9. Electrophilic nitration reaction of Benzaldehyde (C6H5 – CHO) gives:

(A) m-Nitro Benzaldehyde
(B) p-Nitro Benzaldehyde
(C) 2,4 Dinitro Benzaldehyde
(D) o-Nitro Benzaldehyde

Q10. Which of the following alkenes on ozonolysis produces formaldehyde only?

(A) Propene
(B) Methyl propene
(C) 2-Butene
(D) Ethene

Q11. Predict which of the following compound is more basic: (Structures with lone pairs on nitrogen are printed in the source scan.)

(A) CH3–CH2–NH2
(B) CH3–NH–CH3
(C) (CH3)3N
(D) NH3

Q12. In Kolbe-Schmitt reaction, phenol is reacted with:

(A) SiO2 / KOH
(B) SO2 / NaOH
(C) CO2 / NaOH
(D) Br2 / H2O

Q13. An alcohol reacts with PBr3 as 3R–OH + PBr3 → 3R–Br + H3PO3, the reactivity of alcohol will be greater if it is:

(A) A primary alcohol
(B) A secondary alcohol
(C) A tertiary alcohol
(D) CH3 – OH

Q14. Which of the carbonyl compound will NOT undergo aldol condensation reaction with a mild base?

(A) Benzaldehyde
(B) Acetone
(C) Propionaldehyde
(D) Acetaldehyde

Q15. All the acid derivatives can be converted back into corresponding carboxylic acid by their reaction with:

(A) NH3
(B) CH3CH2OH
(C) LiAlH4
(D) H2O

Q16. Which of the following nitrogenous base is NOT present in DNA?

(A) Adenine
(B) Thymine
(C) Uracil
(D) Cytosine

Q17. Nylon –6,6 is the polymer of:

(A) Glycerol and Adipic acid
(B) Hexamethylene diamine and Adipic acid
(C) Hexamethylene diamine and Glycerol
(D) Ethylene Glycol and Adipic acid

> Note: SUPPLEMENTARY TABLE: Atomic No / Symbol / Mass No — 1 H 1; 2 He 4; 3 Li 7; 4 Be 9; 5 B 11; 6 C 12; 7 N 14; 8 O 16; 9 F 19; 10 Ne 20; 11 Na 23; 12 Mg 24; 13 Al 27; 14 Si 28; 15 P 31; 16 S 32; 17 Cl 35.5; 18 Ar 40; 19 K 39; 20 Ca 40.

CHEMISTRY HSSC-II — SECTIONS B & C (Subjective Type)

Time Allowed: 2:35 Hours | **Maximum Marks:** 68 | **Paper Code:** 2HA-I 24005 (B)

> Note: SECTION – B (Marks 42) — Q.2 Answer the following questions briefly. (14x3=42)

(i) [03] Write chemical equations to show what happens when following compounds are heated: a. LiNO3 b. NaNO3 c. Mg(NO3)2 — OR (03) — What is the difference between the functional groups present in the following pairs of compounds? a. Acid halides and Acid amides b. Ethers and Esters c. Aldehydes and Ketones.

(ii) [2+1] Write down the electronic configuration of Copper and Chromium. Why these electronic configurations violate Aufbau principle? — OR (2+1) — Describe the geometries of complex ions with co-ordination number 4 and 6.

(iii) [1+2] State Fajan's rule and justify it by giving two examples. — OR (03) — Write down the reactions of Acetic anhydride with a. H2O b. NH3 c. C2H5OH.

(iv) [03] Justify that the restricted rotation about C-C bond in 1,2-dimethyl cyclo propane is responsible for geometrical isomerism. — OR (03) — How can aldehydes and ketones be differentiated by giving two chemical tests?

(v) [03] Write down the mechanism of SN1 reactions. — OR (03) — Write down three differences between addition and condensation polymerization.

(vi) [1+2] How $Cr_2O_7^{2-}$ and CrO_4^{2-} ions exist in equilibrium with one another? How can they be interconverted due to this equilibrium? — OR (1+2) — Briefly describe enzyme inhibition? What are its types.

(vii) [03] How can 1-Butyne be prepared using a suitable vicinal dihalide and a geminal dihalide? Give chemical reactions with conditions. — OR (1+2) — What is green house effect? Give its brief description.

(viii) [03] Write down the reactions of Phenol with conc. HNO_3, aq. Br_2 and conc. H_2SO_4 — OR (03) — Describe the type of electronic transition that occurs when UV-Vis radiations of 200–800nm wavelength are passed through $CH_2 = CH - CH_2 - OH$.

(ix) [03] Write down three differences between E1 and E2 reactions. — OR (03) — Describe the significance of Functional group region and Finger print regions in the IR spectrum.

(x) [2+1] Justify that $Al(OH)_3$ is amphoteric but $Mg(OH)_2$ is basic. — OR (03) — Write down the reactions of $CH_3 - CH_2 - Mg - Br$ with: a. Acetone b. CO_2.

(xi) [03] Describe anomalous trends in the ionization energies of elements of 3rd period in the periodic table. — OR (03) — Why is Acetaldehyde more reactive than Acetone in nucleophilic addition reactions?

(xii) [2+1] Write down two reactions in which O–H bond of alcohol is broken. What is reactivity order of different alcohols in these reactions? — OR (03) — What is the role of chloro-fluoro carbons (CFC) in destroying ozone layer in the stratosphere?

(xiii) [2+1] Write down the chemical reactions for the following observations: a. Calcium when heated in air (containing O_2 and N_2) results in the formation of two compounds b. One of the above two compounds reacts with water to form a pungent gas. — OR (03) — Naturally occurring magnesium has three isotopes. Mg–24 mass = 24amu %-age abundance = 78.70%; Mg–25 mass = 25amu %-age abundance = 10.13%; Mg–26 mass = 26amu %-age abundance = 11.17%. Calculate relative atomic mass of Mg.

(xiv) [03] How can ethene be converted into: a. Ethanol b. Ethylene chlorohydrin — OR (03) — Acetone undergoes aldol condensation reaction with a mild base. Write down the mechanism of this reaction of acetone.

> Note: SECTION – C (Marks 26) — Note: Attempt the following questions.

Q3. (Marks 1+2+2+2) Q.3 Justify that CO_2 is an acidic oxide. How the dioxides of Ge, Sn and Pb differ in nature from CO_2? Explain by giving two chemical reactions for each in the support of your answer. — OR (1+1+5) — How is the refining of Crude oil carried out? State the basic principle involved and explain the steps in industrial process. (Do not describe the fractions).

Q4. (Marks 3+3) Q.4 Why Carbon and Silicon show +4 oxidation states whereas Ge, Sn and Pb show +4 as well as +2 oxidation states in their compounds? Explain. Also compare the relative stability of +4 and +2 oxidation states in Ge, Sn and Pb. — OR (3+2+1) — Explain the structure of Benzene on the basis of molecular orbital

concept. How does this concept justify the stability of benzene? Why does benzene preferably give electrophilic substitution reactions?

Q5. (Marks 06) Q.5 How can Acetyl Chloride be prepared from carboxylic acid? Write down the reactions of CH3–CO–Cl with: a. Acetic acid b. Ethanol c. NH3 — OR (3+3) — Describe lipids. How can essential and non-essential lipids be differentiated? Write down the following reactions of a general triglyceride: a. Hydrolysis b. Saponification.

Q6. (Marks 1+1+5) Q.6 What is mass spectrometry? Write down its basic principle. Explain the construction and working of a mass spectrometer. — OR (1+2+2+2) — What is the order of acidic strength of phenol, alcohol and carboxylic acid? Justify your answer by giving pKa values, and stability of their conjugate bases.

VERSION D

CHEMISTRY HSSC-II — SECTION – A (Objective Type)

Time Allowed: 25 Minutes | **Maximum Marks:** 17 | **Paper Code:** 42052 (Version D)

Note: Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Note: Fill the relevant bubble against each question according to curriculum:

ہر سوال کے سامنے دیے گئے کریکولم کے مطابق درست دائرہ کو پر کریں۔

Q1. Which of the following alkyne would NOT yield a ketone on hydration reaction?

(A) 2-Butyne
(B) Acetylene
(C) Propene
(D) 1-Butyne

Q2. Which halogen reduces H2SO4 to SO2?

(A) I2
(B) F2
(C) Cl2
(D) Br2

Q3. Li2CO3 is thermally unstable due to:

(A) High charge density of Li+
(B) Low electronegativity of Li+
(C) High electronegativity of Li+
(D) Low charge density of Li+

Q4. Which of the following carbides reacts with water to produce methane gas?

(A) SrC2
(B) Be2C
(C) CaC2
(D) MgC2

Q5. The geometry of complex ion [Cr(en)3]3+ is:

(A) Octahedral
(B) Trigonal planar
(C) Triangular pyramid
(D) Square planar

Q6. Which of the following is NOT obtained as a result of destructive distillation of coal?

(A) Gas oil
(B) Coal gas
(C) Coal tar
(D) Ammonical liquor

Q7. Which of the following compounds will NOT show geometrical isomerism?

(A) Maleic acid
(B) 1-Butene
(C) 2-Butene
(D) Fumaric acid

Q8. Ozonolysis of Ethene gives:

(A) Ethanol
(B) Methanal
(C) Ethanal
(D) Methanol

Q9. The reaction of CH3–CH2–C(=O)–NH2 with Br2 / KOH gives: (Reaction scheme is printed in the source scan.)

(A) CH3–CH2–C(=O)–Br
(B) CH3–CH2–CH2–NH2
(C) CH3–CH2–CH2–Br
(D) CH3–CH2–NH2

Q10. The order of acidic strength of alcohols phenol and carboxylic acids is:

(A) Carboxylic acids > Phenols > Alcohols
(B) Alcohols > Phenols > Carboxylic acids
(C) Carboxylic acids > Alcohols > Phenols
(D) Phenols > Carboxylic acids > Alcohols

Q11. Which of the carbonyl compounds is more reactive in a nucleophilic addition reaction?

(A) Acetone
(B) Methanal
(C) Ethanal
(D) Propanal

Q12. The reaction CH3COOH with LiAlH4 gives:

(A) CH3OH
(B) CH3CH2OH
(C) CH3CHO
(D) CH3OCH3

Q13. Which of the following is NOT a disaccharide?

(A) Galactose
(B) Sucrose
(C) Maltose
(D) Lactose

Q14. Nail polish remover is the mixture of:

(A) Ethyl acetate and carbon disulphide
(B) Benzene and Acetone
(C) Ethyl acetate and acetone
(D) Benzene and Carbon disulphide

Q15. The formation of Ozone hole in stratosphere results in:

(A) Exposure to UV radiation
(B) Acid rain
(C) Global warming
(D) Increase in CO2 conc.

Q16. In UV spectroscopy, which transition requires greater energy?

(A) σ → σ*
(B) n → π*
(C) π → π*
(D) n → σ*

Q17. How many different types of protons are there in Butanoic acid (CH3CH2 – CH2 – COOH)?

(A) 8
(B) 3
(C) 4
(D) 6

> Note: SUPPLEMENTARY TABLE: Atomic No / Symbol / Mass No — 1 H 1; 2 He 4; 3 Li 7; 4 Be 9; 5 B 11; 6 C 12; 7 N 14; 8 O 16; 9 F 19; 10 Ne 20; 11 Na 23; 12 Mg 24; 13 Al 27; 14 Si 28; 15 P 31; 16 S 32; 17 Cl 35.5; 18 Ar 40; 19 K 39; 20 Ca 40.

CHEMISTRY HSSC-II — SECTIONS B & C (Subjective Type)

Time Allowed: 2:35 Hours | **Maximum Marks:** 68 | **Paper Code:** 2HA-I 24005 (D)

> Note: SECTION – B (Marks 42) — Q.2 Answer the following questions briefly. (14x3=42)

(i) [03] Write down the reactions of dil. HCl with Li2O, Na2O2 and KO2 — OR (03) — What is the significance of dissolved oxygen (DO), Biochemical oxygen demand (BOD) and chemical oxygen demand (COD)?

(ii) [2+1] What is the role of effective nuclear charge and number of electronic shells in determining the atomic radii of elements of 3rd period? — OR (03) — MgCl2 is a high melting solid, AlCl3 is a solid that sublimes at 180°C whereas SiCl4 is a volatile liquid. How will you justify this difference in volatility?

(iii) [03] Why Alkali metals impart different colours to the flame? — OR (03) — Write down the mechanism of following reaction: 2CH3 – OH --(Conc.H2SO4 / 140°C)--> CH3 – O – CH3 + H2O.

(iv) [03] How is BaO prepared? Write down chemical equation to show what happens when: a. BaO is heated in air b. Resulting product is reacted with dil H2SO4 — OR (03) — Identify ligands, co-ordination number and geometry of the following complex ion [Co(en)2 Cl2]+.

(v) [03] Write down the reactions of V2O5 with HCl, HNO3, SO2 — OR (03) — Differentiate between atomic emission and atomic absorption spectroscopy.

(vi) [1+2] How is KMnO4 used in redox titration with acidified FeSO4? Describe the procedure with chemical equation. — OR (03) — Differentiate between, secondary and tertiary structures of proteins.

(vii) [2+1] Write down two similarities and one difference between the members of a homologous series. — OR (03) — Briefly describe the process of spin flipping in NMR spectroscopy.

(viii) [03] 1-Propanol shows two types of structural isomerism. Name the types and draw the isomers of each type. — OR (03) — Write down three difference between SN1 and SN2 reaction mechanisms.

(ix) [03] Write down the mechanism for the following reaction of Benzene: Friedel-Crafts acylation of benzene with CH3–CH2–CO–Cl in presence of AlCl3. (Reaction diagram is printed in the source scan.) — OR (03) — What happens when following compounds are reacted with LiAlH4: a. CH3 – CN b. CH3 – CO – OC2H5 c. CH3 – CO – NH2.

(x) [03] How SN2 reaction mechanism can be supported by kinetic and stereochemical evidences? — OR (03) — Write down three adverse effects of dissolved fertilizers present in water as pollutants.

(xi) [03] Write down the reactions of CH3 – CH2 – NH2 with acetyl chloride, Acetaldehyde, HNO2 / HCl — OR (03) — Describe the structural components of nucleotides of RNA.

(xii) [03] How can Dimethyl ether be prepared by: a. Williamson's synthesis b. From an alkyl halide — OR (03) — How can Propanoic acid be prepared from: a. A nitrile b. A Grignard reagent c. An Aldehyde.

(xiii) [03] Write down the reactions of 1-Propanol with. a. SOCl2 b. Conc. H2SO4 /180°C c. K2Cr2O7 / H2SO4 — OR (03) — Write down the composition and uses of following fractions of Petroleum refining. a. Gasoline b. Naphtha.

(xiv) [03] Write down any two applications of iodoform test with chemical equations. — OR (03) — 0.240g of an organic compound contain 0.096g of carbon, 0.016g hydrogen and 0.128g oxygen. Determine empirical formula of the compound.

> Note: SECTION – C (Marks 26) — Note: Attempt the following questions.

Q3. (Marks 3+4) Q.3 Describe resonance and resonance energy. How can the resonance energy of benzene be calculated with the help of heat of hydrogenation? — OR (1+2+4) — Describe polymerization. What are its types? Describe one example of each type with chemical equation for its formation.

Q4. (Marks 1+1+1+3) Q.4 Why halogens act as oxidizing agents? How is the oxidizing property of halogens represented? What is the order of relative power of halogens as oxidizing agent? Prove this order by giving chemical equations. — OR (3+3) — Write down the reactions of Acetaldehyde and Acetone with. a. 2,4-Dinitrophenylhydrazine b. Semicarbazide. (Reaction structures are printed in the source scan.)

Q5. (Marks 3+3) Q.5 Describe the trends in the following physical properties of elements of 3rd period. a. Ionization potential b. Melting point and boiling points — OR (2+2+2) — How can phenol be prepared from: a. Chlorobenzene b. Cumene c. Aniline.

Q6. (Marks 1+2+4) Q.6 What meant by directive influence of substituents on benzene ring? How can substituents be classified on this basis? Give one example of directive influence of substituents for each class with chemical equation. — OR (1+2+2+2+2) — What is IR spectroscopy? Discuss with reference to: a. Reason for absorption of IR-radiations b. IR-spectra c. Application of IR-spectroscopy.